WEEK 1 — MEDIADSN

Introduction to Interactive Media Design

Week 1 — Understanding Interactive Experiences

COURSE	SCOPE	CORE FOCUS	TARGET
MEDIADSN Masterclass	Module 1 • Pages 1 to 36	HCI, Systems, Agency & UI/UX	Foundational Mastery

Course Agenda

What We Will Explore Today

- **Welcome & Core Objectives:** Establishing syllabus targets and studying how digital experiences establish a dynamic two-way conversation with users.
- **Unit 1 – Basics of Interactivity:** Contrast analysis between traditional static media (television, newspapers) and modern user-directed mediums.
- **Unit 2 – The Key Components:** Deconstructing software applications into User, Input, Interface, System, Output, and Feedback elements.
- **Unit 3 – History & Evolution:** Tracing the progress from command-line terminals to modern Graphical User Interfaces (GUIs), touchscreen gestures, and spatial computing.
- **Interactive Lab & Practice:** Interactive simulations (YouTube, Google Maps, ATM), sandboxes, and student assessments.

KEY INSIGHT

Our Goal

By the end of this module, you will understand the fundamentals of interactive media, its historical progression, and be able to audit and identify key components of any interactive system.

Syllabus Learning Objectives

01 Understand the Basics of Interactive Media Design

WHAT IT MEANS

Grasping how digital products establish a dynamic conversation with human users, where inputs directly change outputs.

WHY IT MATTERS

To transition your mindset from static page design (newspapers, layouts) to system design, focusing on user behaviors and feedback states.

STUDENT OUTCOMES

Ability to define interactivity, trace the input-to-response loop, and explain user agency in software.

02 Explore History & Evolution of HCI Models

WHAT IT MEANS

Studying the shift of control models from mechanical boards to command lines, visual desktops (GUIs), web portals, touch, and voice.

WHY IT MATTERS

Understanding past interface transitions helps you predict future interface shifts and design for diverse device form factors.

STUDENT OUTCOMES

Outline major milestones (Engelbart, Xerox PARC, Macintosh) and explain why the GUI model remains a standard visual metaphor.

03 Identify Key Components of Interactive Systems

WHAT IT MEANS

Deconstructing interactive systems into their fundamental parts: User, Interface, Input, System, Output, and Feedback.

WHY IT MATTERS

When debugging user frustration or design failures, breaking the system into parts helps you pinpoint exactly where communication breaks down.

STUDENT OUTCOMES

Audit any software, website, ATM, or mobile application and map all six components clearly.

What Is Interactive Media?

Defining the Conversation

- **Two-way communication:** Unlike traditional mediums, interactive media establishes a dynamic, bidirectional dialogue between a human user and a digital system.
- **User agency and control:** The user has active control over the pace, sequence, and output format of the information, rather than being a passive spectator.
- **Dynamic state updates:** The system receives inputs, executes processing logic, and updates its visual/auditory state in real-time, closing the feedback loop.
- **Cognitive engagement:** Direct engagement increases information retention, mapping to the student's active mental model.

STORY / ANALOGY **Simple Analogy:**

Static media is like a lecture where you just listen. Interactive media is like a conversation where you ask questions, get answers, and decide what to talk about next.

KEY INSIGHT **History Fun Fact: Spacewar! (1962)**

The first truly interactive computer game, "Spacewar!", was created at MIT in 1962. It required real-time coordination on a circular radar-style display screen, marking the birth of real-time software user input loops!

Media vs. Interactive Media

PASSIVE CONSUMPTION

Traditional Static Media

- ✕ Passive Consumption: Reading newspapers, listening to radio broadcasts, or watching motion pictures.
- ✕ Linear Flow: The timeline is pre-recorded. The story flows from start to finish without any user controls.
- ✕ One-Way Delivery: Information is broadcasted; the consumer cannot alter the content in real time.

Core Characteristic: Fixed, linear broadcast. The author dictates pacing and order.

VS

ACTIVE PARTICIPATION

Interactive Media

- ✓ Active Participation: Users choose navigation paths, input custom values, and manipulate visual objects.
- ✓ Non-Linear Exploration: Users can skip around, drill down, search records, or trigger contextual views.
- ✓ Bidirectional Flow: Inputs instantly dictate outputs. The user and the system cooperate.

Core Characteristic: Dynamic bidirectional dialogue. The user directs the exploration.

KEY INSIGHT **The Shift in Responsibility**

In static media, the author controls the exact pacing and linear timeline. In interactive media, the designer builds a system of possibilities, letting the user control the journey.

User Input Simulator

Testing Communication Channels

1. Click / Tap

Pressing physical mouse keys or tapping glass triggers instantaneous coordinate-based action triggers.

Discrete event trigger (mousedown / mouseup)

2. Hover

Positioning indicators inside element boundaries provides proactive visual pre-states before commitment.

Pre-attentive visual cues (tooltips, color changes)

3. Touch Gesture

Direct finger contact on capacitive glass, enabling tactile haptic screen coordinates and pinch-to-zoom.

Multi-touch gestures & pressure sensitivity

4. Swipe

Rapid directional drag across surfaces to paginate carousels, dismiss cards, or reveal contextual drawers.

Velocity & threshold directional vectors

5. Drag & Drop

Continuous pointer holding across spatial coordinates to manipulate virtual objects directly.

Direct object manipulation & bounding cage bounds

6. Type / Keystroke

Alphanumeric keyboard presses compile dynamic string inputs into structured form controls.

High-bandwidth symbolic input stream

INTERACTION TAKEAWAY

Input Modality Defines Human Agency

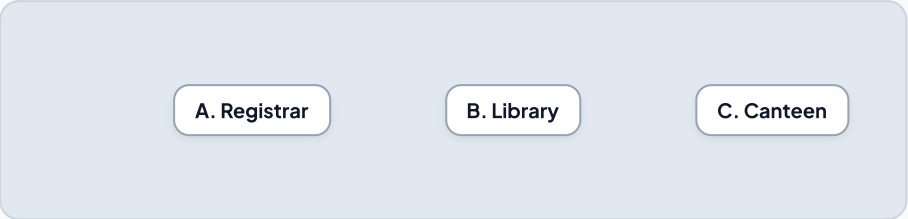
Each input channel carries specific physical affordances and error rates. Great designers match the input method to the user's immediate physical context and task urgency.

Static Mode vs. Interactive Mode

Toggle modes below to experience the psychological difference:

STATIC MODE (PAPER MAP)

Pre-Printed Campus Poster



❌ Fixed information: cannot display current office hours or wait times.
❌ No zoom, search, or real-time GPS location pin.
❌ If hours change, entire batch of physical maps must be re-printed.

INTERACTIVE MODE (DIGITAL GIS)

Dynamic Interactive Campus Map



✓ Real-time status: tap any building to inspect live queue & schedule.
✓ Non-linear exploration: filter by services, search by department.
✓ Bidirectional loop: queries return updated navigation paths instantly.

The Interaction Loop

How Humans and Computers Converse

1 **HUMAN COGNITION**

Goal Formulation

The user defines what they want to achieve in their head (e.g. checking their remaining class attendance score).

→

2 **PHYSICAL ACTION**

Input Execution

The user executes physical actions through coordinates or keys (e.g. clicking the "Grade Report" button).

→

3 **COMPUTATION**

System Processing

The application runs calculations, executes queries, updates database variables, and compiles a new state.

→

4 **PRESENTATION**

Output Display

The system renders visual changes, plays audio prompts, or vibrates to express the new updated state.

→

5 **COGNITIVE CLOSURE**

Evaluation & Feedback

The user perceives feedback, checks if goal was achieved, and decides their next action.

THE GULF OF EXECUTION & EVALUATION

Donald Norman's HCI Foundation

Two fundamental chasms occur in every interaction: **Gulf of Execution** (how hard is it for the user to figure out how to do what they want?) and **Gulf of Evaluation** (how hard is it to tell if the system did what they wanted?). Great design bridges both gulfs smoothly.

Interaction Loop Breakdown

Deep Dive into Each Phase

- **1. Goal Formulation:** The user decides what they want to achieve (e.g., finding the price of a flight).
- **2. Action Execution:** The user interacts with the system using an input device (e.g., clicking the "Search Flights" button).
- **3. Processing:** The system runs algorithms, database queries, and background validations to generate the result.
- **4. Output Representation:** The interface presents the new state (e.g., rendering a list of available flights and price cards).
- **5. Evaluation & Feedback:** The user reads the screen, assesses if their goal was met, and plans their next move.

STORY / ANALOGY

Design Story: The Norman Door

Donald Norman coined the term "Norman Doors" to describe doors with handles that suggest pulling, but actually require pushing. This represents a breakdown in the evaluation loop where the design contradicts user mental models.

Key Components of Interactive Media

The Building Blocks of Digital Experiences

- **The User:** The human operator who has goals, expectations, cognitive limits, and sensory capacities.
- **The Interface:** The digital membrane (screens, audio, physical controls) where interaction occurs.
- **The Input:** The mechanical or gestural action translated by the hardware (clicks, keyboard keys, speech).
- **The System:** The computational hardware and software that computes state changes and processes logic.
- **The Output:** The auditory, visual, or haptic displays returned by the system.
- **The Feedback:** The immediate indicator showing that the system recognized the input and is acting on it.

KEY INSIGHT

💾 History Story: Xerox Alto (1973)

Xerox PARC built the first computer featuring all six modern components, including the first desktop metaphor and three-button mouse. Unfortunately, Xerox corporate executives failed to see its commercial value, allowing Apple and Microsoft to later define the GUI era.

Visualizing Component Connections

Select a component to inspect its relationships and data flows:

1. User

Initiator of Action

The human agent with goals, mental models, emotional context, and physical constraints.

2. Input

Translator of Intent

Physical hardware translating human motor movements into digital coordinates (mouse, keyboard, touch, mic).

3. Interface

Sensory Membrane

The visual layout, typography, controls, and cues that mediate between human and machine.

4. System

Computational Engine

The backend algorithms, database tables, business logic, and memory state that execute commands.

5. Output

State Communication

Visual screens, audio speakers, mechanical actuators, or haptic buzzers presenting system state.

6. Feedback

Loop Closure

Instantaneous confirmation that user action was received, understood, and processed successfully.

SYSTEM ARCHITECTURE

Interconnected Data Flow

User ➔ (acts via) Input ➔ (interpreted by) Interface ➔ (processed by) System ➔ (rendered via) Output ➔ (perceived as) Feedback ➔ User. If any single link fails, the user experiences confusion, error, or abandons the application.

Case Study: YouTube Video Player

Mapping Components to YouTube Controls

- **User Intention:** Find and watch an instructional design video tutorial.
- **Interface controls:** Buttons for play/pause, dynamic progress bars, settings menus, and volume sliders.
- **Input translation:** Mouse clicks, touchscreen gestures, or key taps register control commands.
- **System Processing:** Network servers seek video timestamps, buffer segments, and calculate recommendations.
- **Feedback & Output:** Video starts immediately, progress bar advances, and buffer indicators spin during delays.

STORY / ANALOGY **Play with the simulator:**

Adjust options on the right simulation widget. Notice how system inputs instantly alter video output and status updates!

Case Study: Google Maps Navigation

Exploring Non-Linear Geographic Viewports

- **User Intention:** Find the fastest route to the campus registrar office.
- **Interface controls:** Map overlays, search inputs, zoom widgets, and directions buttons.
- **Input translation:** Screen swipes to pan, pinch to zoom, and keystrokes to input search strings.
- **System Processing:** Routing engines query database tables and calculate path directions.
- **Feedback & Output:** Route paths trace on the canvas, compass pointers rotate, and directions list updates.

STORY / ANALOGY **Play with the maps:**

Type destinations or tap buttons in the maps simulator on the right. Notice how the map adjusts views dynamically!

Case Study: Automated Teller Machine

Rigid Security Interaction Loops

- **User Intention:** Safely withdraw cash from checking balance records.
- **Interface controls:** Numeric pinpads, card intake slots, and cash dispenser gates.
- **Input translation:** Keypresses to input PIN numbers, button taps to select withdrawal cash values.
- **System Processing:** Banking servers authenticate credentials, balance database records, and verify currency notes.
- **Feedback & Output:** Card reader locks, status indicators prompt actions, dispenser ejects banknotes.

STORY / ANALOGY **Try the ATM:**

Complete the secure login PIN sequence on the simulator to withdraw cash and trigger state feedback.

History and Evolution of Interactive Media

From Scientific Tools to Personal Expression

- **Early Computers (1940s–50s):** Giant machines (like ENIAC) configured via physical cables and punch cards. Zero real-time user interaction.
- **Command-Line Interfaces (1960s–70s):** Teletypes and terminals allowed text commands. Interaction was slow, sequential, and text-only.
- **The GUI Revolution (1980s):** The mouse, folders, icons, and windows (WIMP model) made computing visual and accessible.
- **The World Wide Web (1990s):** Hyperlinks connected documents, introducing non-linear reading across global networks.
- **Mobile & Touch (2000s):** Capacitive screens replaced styluses and mice with direct, finger-based touch gestures (pinch, swipe).
- **Immersive Media (Modern):** Virtual Reality (VR), Augmented Reality (AR), and voice assistants remove physical screen barriers.

KEY INSIGHT

💡 History Story: "Mother of All Demos" (1968)

In 1968, Douglas Engelbart presented the computer mouse, hypertext links, dynamic file browsing, and video conferencing in a single demonstration. It took the commercial market nearly 20 years to catch up and implement what he showed in one afternoon!

Interactive Evolution Timeline

Click the milestones below to explore the shift in interaction models:

1950s - 1960s

TEXT / CLI

Batch & Command Line

Punch cards and teletype command terminals. Interaction was rigid, non-visual, and required memorizing arcane syntax strings.

```
C:> RUN ANALYZE.BAT
```

1970s - 1984

WIMP / GUI

Direct Manipulation

Xerox Alto and Apple Macintosh introduce Windows, Icons, Menus, and Pointer (WIMP). The desktop metaphor lets anyone point and click.

```
Folder (Double-Click) ➔ Open
```

1990s - 2000s

HYPERTEXT / WEB

The World Wide Web

Tim Berners-Lee introduces HTML & hyperlinks. Non-linear global exploration across interconnected hypermedia documents.

```
<a href="...">Click to Navigate</a>
```

2007 - 2010s

CAPACITIVE TOUCH

Mobile Gestures

iPhone and capacitive glass remove the mouse barrier. Direct finger touch, multi-touch pinch, swipes, and inertia scrolling.

```
Swipe Up ➔ App Switcher
```

2020s & Beyond

SPATIAL & AI

Spatial & Conversational

Eye-tracking, hand-tracking spatial computing (Vision Pro, Meta Quest) and natural language generative AI copilots.

```
Pinch in air + Voice Command
```

THE CORE TREND

From Machine-Centric to Human-Centric

Every interface evolution lowered the cognitive barrier: from humans learning machine code ➔ humans typing terminal syntax ➔ humans recognizing visual real-world metaphors ➔ humans directly touching virtual glass ➔ machines understanding natural human voice and intent.

Command Line vs. Graphical User Interface

Try executing the task "Open Document" in both environments:

TEXT ENVIRONMENT

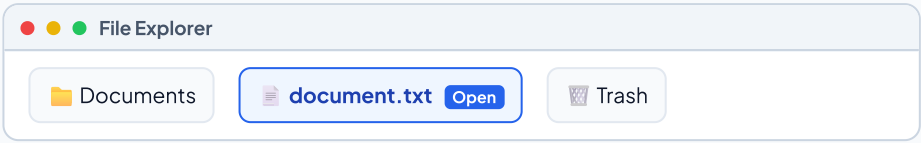
Command Line Interface (CLI)

```
bash — 80x24
$ ls -la
drwxr-xr-x docs
-rw-r--r-- document.txt 4.2KB
$ cat document.txt
Interactive Media Lecture Notes loaded.
```

- Requires **Recall**: Users must memorize exact command names and arguments.
- High error penalty: A typo like `rm -rf /` can erase entire systems.
- Incredibly fast and automatable for experienced engineers.

VISUAL ENVIRONMENT

Graphical User Interface (GUI)



- Relies on **Recognition**: Visual icons indicate affordances clearly.
- High error tolerance: Dragging to Trash can be easily undone with Cmd+Z.
- Zero prerequisite memorization: Intuitive for children and non-technical users.

Why the GUI Succeeded

The Power of Visual Metaphors

- **Desktop Metaphor:** The screen resembles a physical desk with folders, files, and trash bins, tapping into real-world familiarity.
- **Direct Manipulation:** Instead of typing abstract text instructions, users grab, drag, and drop visual files directly.
- **Recognition over Recall:** Command lines require users to memorize exact commands (recall). GUIs present icons and menus (recognition).
- **Immediate Visibility:** The system changes state immediately upon interaction, showing progress clearly.

CORE CONTEXT

Module Focus

This principle forms the cornerstone of user engagement, predictable cognitive mental models, and robust system architecture in modern interactive software.

Major Categories of Interactive Media

Diverse Formats and Scenarios

- **Interactive Websites:** Portals like E-Commerce shops and blogs where users search, filter, check out, and chat.
- **Video Games:** Complex, highly interactive simulations where players control narratives, actions, and character parameters.
- **Mobile Applications:** Handheld touch tools optimized for speed, gestures, location mapping, and haptics.
- **Digital Kiosks:** Public terminals (airports, retail, museums) with touch-optimized maps or ticketing.
- **Interactive Art Installations:** Spaces where sensors detect motion, sound, or temperature, causing projection updates.

STORY / ANALOGY **Key Point:**

Regardless of the category, the fundamental rule remains: design must prioritize user-centered navigation and immediate, clear response loops.

Affordances, Signifiers, and Feedback

Helping Users Understand the System

- **Affordance:** The actual physical/digital properties of an object that define how it can be used (e.g., a button affords clicking).
- **Signifier:** Visual indicators that point out the affordance (e.g., a drop shadow and rounded corners signaling that it is a button).
- **Feedback:** What happens after the action (e.g., a clicking sound, button press animation, and page transition).
- **Mapping:** The natural relationship between controls and their movements (e.g., sliding a volume knob right increases intensity).

STORY / ANALOGY

History Fact: Jobs and the Stylus

When Apple introduced the iPhone in 2007, Steve Jobs famously declared: "Who wants a stylus? You have to get 'em, put 'em away, you lose 'em... Yuck! We're going to use the best pointing device in the world - our fingers." This marked the transition to pure capacitive gestures!

Good Interaction vs. Poor Interaction

PASSIVE CONSUMPTION

Good Interaction Design

- ✗ Clear visual signifiers: A button clearly stands out from static text layout elements.
- ✗ Immediate feedback: System displays a spinner or visual change in under 100ms.
- ✗ Informative labels: Buttons read "Complete Purchase" or "Delete Folder".
- ✗ Forgiving controls: Provides "Undo" and clear confirmation modals before destruction.

Core Characteristic: Fixed, linear broadcast. The author dictates pacing and order.

VS

ACTIVE PARTICIPATION

Poor Interaction Design

- ✓ Invisible buttons: Interactive elements look exactly like static bullet points.
- ✓ No confirmation: Tapping "Format Drive" deletes files instantly with no warning.
- ✓ Vague actions: Buttons are labeled "Submit" or "Click Here" without context.
- ✓ Zero status feedback: Clicked button does not animate, leaving users guessing.

Core Characteristic: Dynamic bidirectional dialogue. The user directs the exploration.

KEY INSIGHT

⏱️ Design Story: The Progress Bar (1985)

In 1985, Brad Myers presented the concept of a progress indicator at ACM SIGCHI. He proved that showing a loading bar reduces user anxiety and prevents them from restarting applications thinking the computer has crashed.

Interactive Button State Sandbox

Inspect and click the control below to see state updates:

1. DEFAULT / NORMAL

Submit Application

Resting baseline state. Clear label, high contrast border, and drop shadow signaling clickable affordance.

2. HOVER (:hover)

Submit Application

Pointer detected inside boundaries. Background darkens/lightens; cursor switches to pointer hand.

3. FOCUS (:focus-visible)

Submit Application

Keyboard tab navigation indicator. High-contrast outline ring ensuring WCAG accessibility compliance.

4. ACTIVE / PRESSED

Submit Application

Physical mouse button down or touch contact. 95% scale reduction and shadow collapse mimicking physical depression.

5. LOADING / IN-FLIGHT

↻ Processing...

Prevents duplicate double-clicks. Replaces text with spinner to reassure user that request is processing.

6. DISABLED / MUTED

Submit Application

Form requirements incomplete. Muted opacity (40%), not-allowed cursor, removed from tab focus ring.

UX ACCESSIBILITY MANDATE

Every Interactive Control Must Have All 6 States

Never design a button that only has a default state. Without hover, focus, active, and loading indicators, users will repeatedly click and wonder if the system has frozen.

Levels of Interactivity Complexity

Slide the slider to inspect responsive layouts from static up to simulator levels:

LEVEL 1

Passive / Static

User is a pure reader or viewer. Zero state updates.

- PDF Handout
- Blog Article
- Billboards

LEVEL 2

Reactive Navigation

User clicks links, tabs, or accordions to reveal sections.

- Documentation Wiki
- Menu Dropdowns
- FAQ Accordions

LEVEL 3

Bidirectional Manipulative

User enters data, drags components, filters records in real time.

- E-Commerce Filters
- Canva Canvas
- Spreadsheets

LEVEL 4

Adaptive / Generative / Simulated

Complex simulations, physics engines, and AI agents adjusting to user behavior.

- 3D CAD / Figma
- Flight Simulators
- Video Games

DESIGN DECISION MATRIX

Matching Complexity to User Context

Higher interactivity is not automatically better. An emergency medical alert needs Level 1 clarity, while a creative workstation needs Level 4 freedom. Design for the user's cognitive load.

System Responsiveness Timelines

Understanding Interaction Response Thresholds

- **0.1 Second (100ms):** The limit for users to feel that the system is reacting instantaneously. Visual feedback (button states) must occur within this window.
- **1.0 Second:** The limit for the user's flow of thought to stay uninterrupted. If actions take longer, loading indicators must show.
- **10 Seconds:** The limit for keeping the user's attention focused. Actions taking longer require progress indicators showing completion time.

KEY INSIGHT

Miller's Law

A user's short-term memory can hold 7 ± 2 chunks of info. If response times exceed 10 seconds, their goal memory decays, breaking the loop.

User Control and Freedom

Providing Emergency Exits

- **Undo/Redo:** Giving users simple commands to cancel accidental actions or go back safely.
- **Emergency Exits:** Clear "Cancel", "Back", or "Quit" buttons on multi-step forms so users do not feel trapped.
- **Configurable Defaults:** Letting users customize their settings and revert parameters easily.
- **Forgiving systems:** Encouraging exploration by making actions non-destructive where possible.

STORY / ANALOGY

⌨️ History Story: The Origin of Undo (1971)

The "Undo" command was created by programmer Warren Teitelman in 1971 while working on the INTERLISP system at Xerox PARC. It was originally called "UNDO" and was designed to save programmers from losing hours of code due to typo mistakes.

User Agency vs. Machine Automation

Balancing Automation and Control

- **User Agency:** Giving the student or operator full control over parameters, letting them adjust styles and settings.
- **System Automation:** Automating tedious operations (saving data, validating variables) to prevent cognitive overload.
- **Avoid Trap Gates:** Designers must never take away navigation freedom unless security parameters mandate locking inputs.
- **Informative Interactivity:** Interaction should clarify concepts, not confuse users with unnecessary clicks.

KEY INSIGHT **Design Philosophy**

The machine should assist, not dictate. High-agency designs empower users to configure states, building confidence.

Case Study: Campus Information Kiosk

Designing for Public Interaction Scenarios

- **The Scenario:** Trimex Colleges wants to deploy public touchscreen kiosks in lobby hallways.
- **The Goal:** Help freshmen locate the Registrar Office, verify today's events, and check classroom schedules.
- **The Challenge:** Users are in a hurry, have different touch heights, and expect immediate feedback without reading manual text pages.

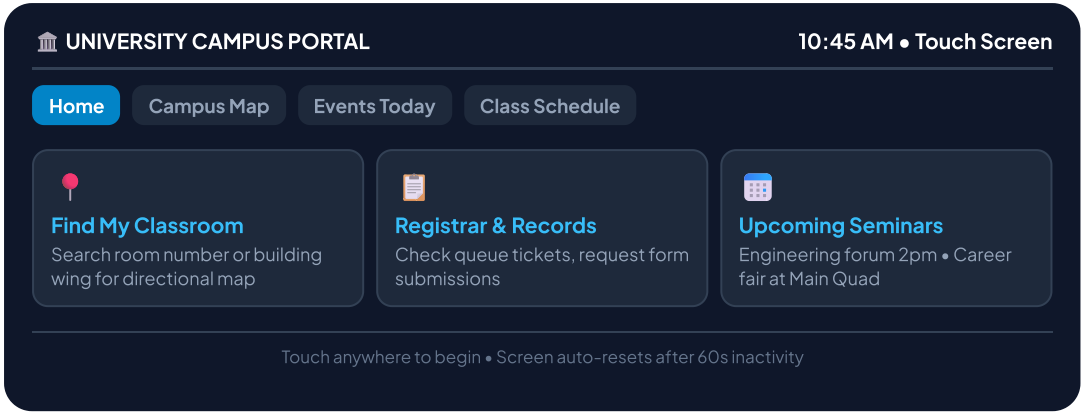
STORY / ANALOGY

Design Goal:

We must build an interface with large touch zones, visible signifiers, and instant visual response maps to facilitate low-friction interaction loops.

Interactive Lobby Kiosk Mockup

Tap portals on screen below to check Registrar directions and events calendars:



Public Kiosk Interaction Rules

Huge Touch Targets: Buttons must measure at least 48×48px (ideally 72px+) for finger tapping without precision stylus.

High Ambient Contrast: Screens must remain legible in bright sunlit university lobbies and glare.

Automatic Session Timeout: Auto-resets to Welcome Screen after 60 seconds of inactivity to protect student privacy.

Physical Accessibility: Interface controls placed between 36" and 48" from ground for wheelchair accessibility (ADA/Bautista guidelines).

Activity: Chronological Interaction Loop

Sort items on left in correct chronological order to complete the loops:

✓ VERIFIED MASTER SEQUENCE

The 5-Step Interaction Loop Sequence

STEP 1 **User Goal (Intent)**
The human formulates a mental objective (e.g., wanting to send an urgent email to their professor).

STEP 2 **Input Action (Execution)**
The user presses physical mouse buttons, touches screen coordinates, or types keystrokes.

STEP 3 **System Processing (Logic)**
The computational hardware parses instructions, triggers database queries, and updates app state.

STEP 4 **Output Display (Render)**
The display panel renders visual pixels, plays sound effects, or triggers haptic vibrations.

STEP 5 **Feedback (Evaluation & Closure)**
The user perceives the visual confirmation, checks if the email was sent, and concludes the mental task.

The User-Centered Design Process (UCD)

Structuring Your Creative Design Framework

- **1. User Research:** Understand user demographics, cognitive capabilities, and target environments.
- **2. Requirements Definition:** Specify user goals and technical limitations.
- **3. Interface Prototyping:** Build visual layouts, interactive wireframes, and mockups.
- **4. Usability Evaluation:** Test prototypes with actual users to uncover navigation friction.

KEY INSIGHT

🖱️ History Story: The First Computer Mouse (1964)

In 1964, Douglas Engelbart built the first mouse prototype out of a hollowed block of wood, featuring two metal wheels and a single red button. He tested several other models (including foot-controlled ones!), but found the hand mouse was the most natural pointing metaphor.

Knowledge Check: Interactivity

QUESTION EVALUATION

Identify the core definition of interactivity:

A. High-resolution imagery displaying complex visual design.

B. Bidirectional information flow allowing user agency to alter system state. ✓ Correct Answer

C. Processor speeds compiling server pages in less than 50ms.

D. The capacity to print layout grids to paper sheets.

💡 Explanatory Rationale

Correct! Interactivity is fundamentally characterized by a dynamic two-way information loop and the user's agency to alter the system state.

DISTRACTOR ANALYSIS:

Option A: High-resolution images are static media formatting values, not indicators of dynamic interactivity.

Option C: Fast rendering speed is a performance metric, not a measure of interactive capabilities.

Option D: Printing is a physical output channel, but it does not represent dynamic two-way communication.

Knowledge Check: Visual Feedback

QUESTION EVALUATION

A loading spinner spinning on form submit represents which component:

A. Input execution coordinates tracking cursor lines.

B. Backend system compilation running on database servers.

C. **Visual feedback validating action registration to reduce user uncertainty.** ✓ Correct Answer

D. Goal formulation values inside the student's memory.

💡 Explanatory Rationale

Correct! The loading spinner is visual feedback reassuring the user that the system registered the action and is processing.

DISTRACTOR ANALYSIS:

Option A: Input represents user actions (clicks/keypresses) translating goals, not a system display.

Option B: Processing represents computing database changes behind the scenes (invisible).

Option D: Goal formulation is the intention inside the human's head, not a screen output.

Knowledge Check: Evolution Timeline

QUESTION EVALUATION

What is the correct chronological timeline of HCI evolution models:

A. Keyboard typing → Touch screens → Command-Line → Graphical GUI desktop.

B. Punch card boards → Command-Line Terminals → Visual Desktop GUIs → Touch Gestures. ✓ Correct Answer

C. Capacitive touch screens → Desktop folders → Teletypes → Mechanical punch cards.

D. Keyboard CLI → Punch cards → Immersive AR → Hypertext web links.

💡 Explanatory Rationale

Correct! Computing progressed chronologically from physical punch cards to command-line terminals, visual GUIs, and mobile touch gestures.

DISTRACTOR ANALYSIS:

Option A: Keyboard typing and punch cards are in the wrong order; CLI came before GUI.

Option C: This order is reversed; touch interfaces are modern, while punch cards are mid-20th century.

Option D: Keyboard typing came after punch cards, and CLI came before GUI.

Written Essay Assessment

STUDENT WRITTEN AUDIT

Written Essay Assessment Prompt

> **Prompt:** Explain how interactive media has changed the way users interact with digital systems. Discuss at least one real-world example (e.g., YouTube, Canvas, ATM, Spotify) and identify the six core components (User, Input, Interface, System, Output, Feedback) involved.

Submission Writing Guidelines:

- **Target Word Count:** 250 - 400 words.
- **Structure:** Introduction (Definition & Shift) ➔ Real-World Audit (6 Components Breakdown) ➔ Conclusion (Key Lessons).
- **Format:** Clear analytical paragraphs with technical terminology used accurately.

EVALUATION RUBRIC (100 PTS)

Grading Criteria Breakdown

1. Core Definition (30%)
Thoroughly explains the bidirectional interaction loop and contrasts active agency against passive static broadcast media.

2. History Connection (20%)
Accurately references the shift from CLI text commands to GUI visual desktop metaphors and modern touchscreens.

3. Components Audit (30%)
Accurately maps all six loop elements (User, Input, Interface, System, Output, Feedback) on the selected case study.

4. Composition & Clarity (20%)
Writing is grammatically clean, logically organized, and demonstrates professional academic synthesis.

Exit Slip Reflection Log

COGNITIVE CLOSURE

Exit Slip Reflection Activity

Before concluding Module 1, reflect on your daily interactions with digital software and evaluate your comprehension:

1. **Real-World Affordances:**
   Think of an app you use every day (Instagram, Spotify, Canvas). What is one button or control where the signifier made its affordance instantly obvious without instructions?

2. **Gulf of Evaluation:**
   Recall a software bug or frustrating app you experienced recently. Did it freeze without a loading spinner? Which of the 6 components broke the feedback loop?

3. **Key Takeaway:**
   In your own words, what is the single biggest difference between designing a static poster versus designing an interactive web application?

Module Self-Assessment Checklist

- ☑ Can define interactivity vs static broadcasting
- ☑ Can draw and label the 5-step Interaction Loop
- ☑ Can identify User, Input, Interface, System, Output, Feedback
- ☑ Can explain why the GUI desktop model replaced CLI
- ☑ Can describe affordances, signifiers, and feedback rules

Next Up in Week 2: UI Visual Hierarchy, the 60–30–10 Color Rule, WCAG Accessibility, and Nielsen's 10 Usability Heuristics!

Module Summary Roadmap

Key Takeaways from Week 1

- **Interactivity is a bidirectional loop:** User goal → Input → System processing → Output display → Feedback loop.
- **HCI has evolved:** Command lines required recall; GUIs introduced WIMP recognition; touch made coordinates direct.
- **Good design signposts signifiers:** Helping users understand affordances with fast responsiveness grids.

KEY INSIGHT

Keep Exploring!

In Week 2, we will dive into Desktop Visual Metaphors and learn how window layering systems manage multi-process rendering contexts!